

ANUJ SINGH RANA

Backend/Full-Stack Developer

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GitHub: github.com/Anuj8171

PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in backend development with expertise in Node.js, Python, REST APIs, databases, and system integration. Built production-grade projects including sync automation, encrypted chat, and security tools. Strong foundation in DSA and full-stack development. Seeking SDE/Backend Developer roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, SQL, HTML, CSS

Backend: Node.js, Express.js, REST API, Socket.io, Hono

Frontend: React.js, Next.js, TailwindCSS, Zustand

Database: PostgreSQL, MongoDB, SQLite, Prisma, Supabase

Security: OAuth 2.0, JWT, bcrypt, AES-256, TweetNaCl, End-to-End Encryption

Tools: Git, GitHub, Cloudflare, Pyairtable, Arcjet, Connection Pooling

PROJECTS

Two-Way Sync Automation System — Python, REST API, SQLite, Pyairtable

- Engineered bi-directional data synchronization between Trello and Airtable using REST APIs and Pyairtable for automated CRUD operations
- Implemented SQLite database with connection pooling for state management, conflict resolution, and data integrity
- Developed polling mechanism and webhooks to detect and propagate real-time changes across platforms
- Built error handling with retry logic, idempotency checks, and structured logging for production reliability

End-to-End Encrypted Chat Application — React, Node.js, Express.js, Socket.io

- Built secure real-time messaging platform with TweetNaCl end-to-end encryption and secure key exchange protocol
- Implemented OAuth 2.0 and bcrypt authentication with Arcjet rate limiting for platform security
- Developed React frontend with Zustand state management and Express.js backend with Socket.io for bidirectional communication

Secure File Deletion Application — Python, Tkinter, AES-256, Cryptography

- Developed GUI-based file deletion system with DoD 5220.22-M compliant 3-pass and 7-pass overwrite algorithms
- Implemented AES-256 cryptographic erasure with secure key destruction ensuring permanent data irrecoverability
- Designed Tkinter interface with file/folder selection, real-time progress tracking, and detailed operation logs

EDUCATION

Bachelor of Technology in Computer Science

Aug 2022 – Present

Graphic Era University, Dehradun — CGPA: 7.8/10

Higher Secondary (12th), C.B.S.E

2022

Army Public School — 79.45%

Secondary (10th), I.C.S.E

2020

Masseeh Dilasa School — 77.3%

ACHIEVEMENTS & CERTIFICATIONS

- Completed Salesforce Developer Program with AgentBlazer Champion recognition
- Active competitive programmer practicing Data Structures and Algorithms on coding platforms